

12-Week Data Scientist Study Tracker

Month 1: Foundations

Week 1 - Python for Data Science

- Learn NumPy: arrays, indexing, math ops
- Learn Pandas: DataFrames, Series, groupby
- Learn visualization: Matplotlib & Seaborn
- Mini Project: Titanic Dataset EDA

Week 2 - Statistics & Probability

- Descriptive stats: mean, median, std dev
- Probability: distributions, Bayes theorem
- Hypothesis testing: p-value, t-test
- Practice on real datasets

Week 3 - Exploratory Data Analysis (EDA)

- Handling missing values, outliers
- Distributions, groupby, aggregations
- Heatmap, scatter, boxplot visualizations
- Mini Project: Housing price EDA

Week 4 - SQL for Data Science

- Learn SELECT, WHERE, JOIN, GROUP BY
- Practice on Hackerrank / Mode
- Mini Project: SQL-based sales dashboard

Month 2: Machine Learning Core

Week 5 - Machine Learning Basics

- Learn supervised ML: regression, classification
- Train/test split, evaluation metrics
- Mini Project: Linear regression model

Week 6 - Tree-based Models

- Decision Trees and Random Forest

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- Confusion matrix, precision, recall, ROC-AUC
- Project: Churn prediction

Week 7 - Unsupervised Learning

- Clustering: KMeans, DBSCAN
- Dimensionality Reduction: PCA
- Project: Customer segmentation

Week 8 - Feature Engineering & Pipelines

- Scaling, encoding, feature selection
- Sklearn pipelines, GridSearchCV
- Project: End-to-end ML pipeline

Month 3: Real-World Projects & Job Prep

Week 9 - Deep Learning Basics (Optional)

- Understand neural nets, activations
- Build basic model in TensorFlow/Keras
- Project: MNIST digit classification

Week 10 - Deployment & Git

- Use GitHub to host code
- Build Flask/Streamlit app
- Project: Deploy ML model to Render/Heroku

Week 11 - Resume & Portfolio

- Build resume, update GitHub & LinkedIn
- Add 3+ projects with READMEs
- Write Medium blog on 1 project

Week 12 - Interview Prep

- Revise Python, Stats, ML Q&A
- Do mock interviews
- Apply to 10+ internships/jobs

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